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GitHub Link	https://github.com/192425245/ml0403

SIMATS ENGINEERING

CO2 - ASSESSMENT TOOL 2

Scenario-Based Questions

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO 2:Students will be able to apply supervised learning algorithms such as linear regression, classification models, decision trees, neural networks, support vector machines, and ensemble methods to solve real-world prediction and classification problems. They will gain the ability to select appropriate models, train them using datasets, and evaluate their performance. Students will also understand the strengths and limitations of different supervised learning approaches.(BTL-4)

CO Weightage : 15%

Assessment Tool Weightage: 40%

Duration: 90 minutes

Marks: 25

Code of Conduct:

I K. Neeha(192425245) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

K.Neeha

Signature of the student:

Name of the student:K.Neeha

Scenario-Based Questions

CO-AT2

QUESTIONS: 05

Total Marks:25

1.Unsupervised Training of Neural Networks, Auto Encoders

A hospital wants to identify abnormal ECG patterns from thousands of unlabeled records. Which deep learning technique from the syllabus would you recommend? Justify your choice and explain the workflow.

Problem Statement

Modern hospitals generate thousands of Electrocardiogram (ECG) recordings every day through patient monitoring systems, wearable devices, and diagnostic equipment. Manually analyzing each ECG signal is a time-consuming process that requires expert cardiologists. Since most of these records are unlabeled, traditional supervised learning techniques cannot be applied effectively. Therefore, an intelligent system is required to automatically identify abnormal heart patterns, reduce diagnosis time, and assist doctors in making accurate clinical decisions.

Proposed Deep Learning Technique

An Autoencoder is an unsupervised deep learning model that learns the underlying structure of normal ECG signals without requiring labeled data. It consists of two main components: an Encoder, which compresses the input ECG signal into a lower-dimensional representation, and a Decoder, which reconstructs the original signal from the compressed representation. During training, the model minimizes the reconstruction error for normal ECG patterns. When abnormal ECG signals are provided, the reconstruction error becomes significantly higher because the model has not learned those abnormal patterns. This difference in reconstruction error is used to identify possible cardiac abnormalities.

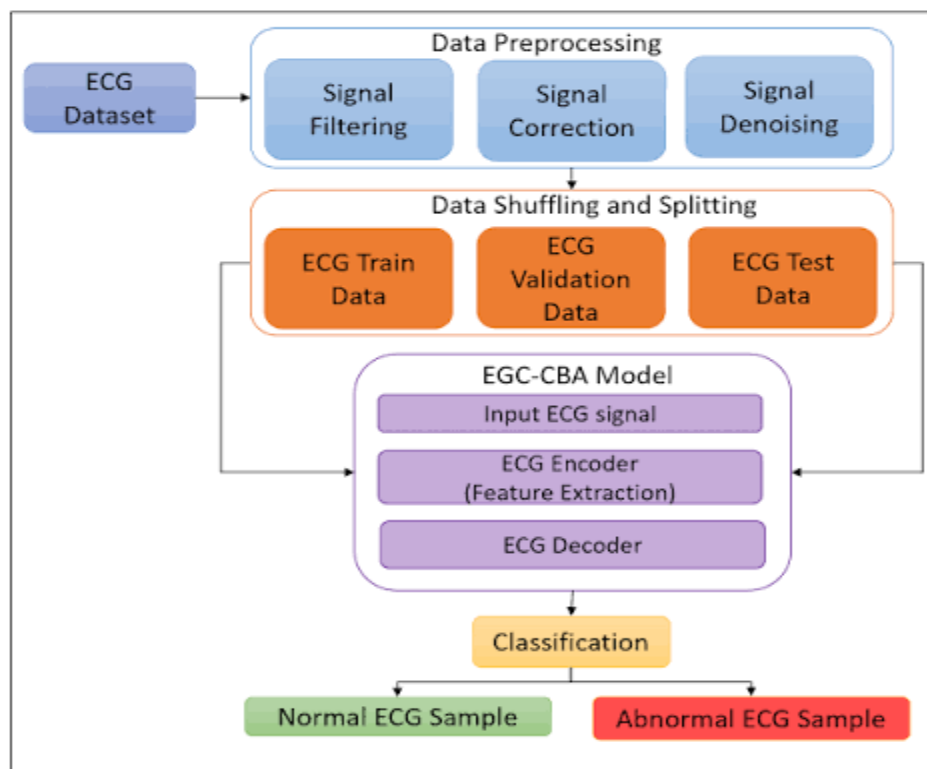
System Architecture

- ECG Signal Collection
- Data Preprocessing and Noise Removal
- Encoder Network
- Latent Feature Representation
- Decoder Network
- Signal Reconstruction
- Reconstruction Error Calculation
- Abnormality Detection and Medical Alert

Working Process

1. Collect ECG recordings from patient monitoring systems.
2. Remove noise and normalize ECG signals.
3. Split the dataset into training and testing sets.
4. Train the autoencoder using only normal ECG recordings.
5. Encode ECG signals into compact latent features.
6. Decode the compressed features to reconstruct the original ECG.
7. Measure the reconstruction error using Mean Squared Error (MSE).
8. If the error exceeds a predefined threshold, classify the ECG as abnormal.
9. Generate an alert for medical experts to perform further diagnosis.

FIGURE:1



Explanation of the ECG Autoencoder Model

Overview

The figure represents a deep learning framework used to automatically detect abnormal ECG signals. The system uses an **Autoencoder**, an unsupervised neural network, to learn the normal characteristics of ECG signals. After training, it can identify abnormal heartbeats by analyzing how well the ECG signal is reconstructed.

Step 1: ECG Dataset

The process begins with an **ECG Dataset**, which contains electrical signals generated by the human heart. These signals are collected from ECG machines, wearable devices, or hospital monitoring systems.

Purpose:

- Provide input data for model training and testing.
 - Include thousands of ECG recordings collected from different patients.
-

Step 2: Data Preprocessing

Before training the neural network, the ECG signals undergo preprocessing to improve their quality.

Q2. Customer Purchase Analysis Using Autoencoders

Problem Statement

Retail companies collect enormous amounts of customer purchase information from online shopping platforms, supermarkets, and mobile applications. This data includes customer demographics, purchase frequency, product categories, payment methods, seasonal trends, and transaction history. As the number of features increases, analyzing customer behavior becomes more difficult due to the high dimensionality of the dataset. Therefore, an efficient dimensionality reduction technique is required before performing customer segmentation.

Proposed Solution

An **Autoencoder** is used to reduce the dimensionality of customer purchase data by learning meaningful compressed representations. Unlike traditional dimensionality reduction methods such as PCA, autoencoders can capture complex nonlinear relationships among customer features. The compressed latent representation generated by the encoder preserves the most important purchasing patterns while removing redundant information.

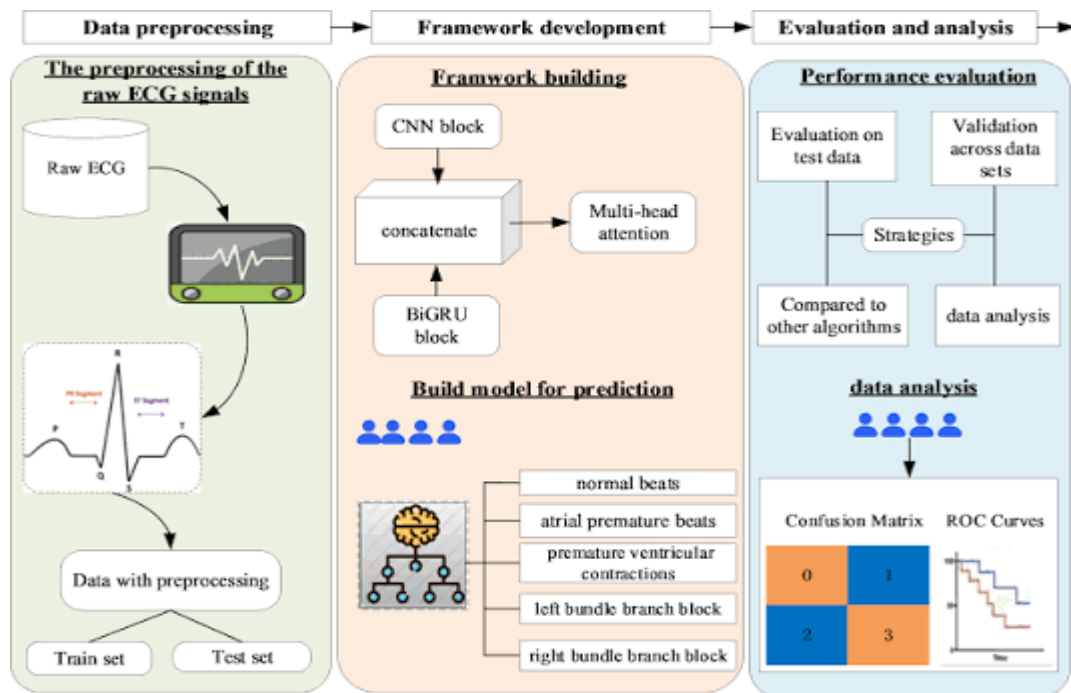
System Workflow

- Collect customer transaction data.
- Remove duplicate and missing records.
- Normalize numerical attributes.
- Train the encoder-decoder architecture.
- Learn compressed customer representations.

- Extract latent feature vectors.
- Apply clustering algorithms such as K-Means or DBSCAN.
- Generate customer segments for marketing strategies.

Business Benefits

- Faster customer segmentation.
- Reduced computational complexity.
- Better understanding of customer purchasing behavior.
- Improved personalized product recommendations.
- Increased customer satisfaction.
- Better inventory and sales forecasting.
- More effective targeted advertising campaigns.



Applications

- E-commerce recommendation systems
- Customer loyalty analysis
- Market basket analysis
- Personalized marketing
- Business intelligence

Expected Result

The autoencoder successfully transforms high-dimensional purchase data into compact feature representations that improve clustering quality. Retail companies can identify valuable customer groups and design personalized marketing campaigns that increase customer retention and revenue.

Signal Filtering

Filtering removes unwanted frequencies and interference such as power-line noise and baseline drift.

Purpose:

- Improve signal quality.
- Remove external electrical disturbances.

Signal Correction

Corrects distortions and aligns ECG waveforms so that all signals have a consistent format.

Purpose:

- Normalize signal amplitude.

Advantages

- Does not require manually labeled ECG datasets.
- Detects unknown and rare cardiac abnormalities.
- Supports continuous real-time patient monitoring.
- Reduces the workload of cardiologists.
- Provides faster diagnosis and early disease detection.
- Improves healthcare efficiency and patient safety.

Applications

- ICU patient monitoring
- Wearable health devices
- Remote healthcare systems
- Smart hospitals
- Cardiac disease screening

Expected Outcome

The proposed autoencoder-based system accurately distinguishes between normal and abnormal ECG signals by analyzing reconstruction errors. This enables early detection of cardiovascular diseases, reduces diagnostic delays, improves patient care, and supports doctors in making timely medical decisions.

3.Width and Depth of Neural Networks, Activation Functions

A facial recognition system performs poorly when trained using a shallow neural network. Recommend architectural changes involving network depth and activation functions.

Problem Statement

A facial recognition system developed using a shallow neural network performs poorly when recognizing faces captured under different lighting conditions, facial expressions, camera angles, and image quality. The limited depth of the network restricts its ability to learn complex facial features, resulting in lower recognition accuracy.

Recommended Architecture

To improve performance, the shallow neural network should be replaced with a **Deep Neural Network (DNN)** containing multiple hidden layers. The deeper architecture enables the network to learn hierarchical facial representations, starting from simple edges to complex facial structures.

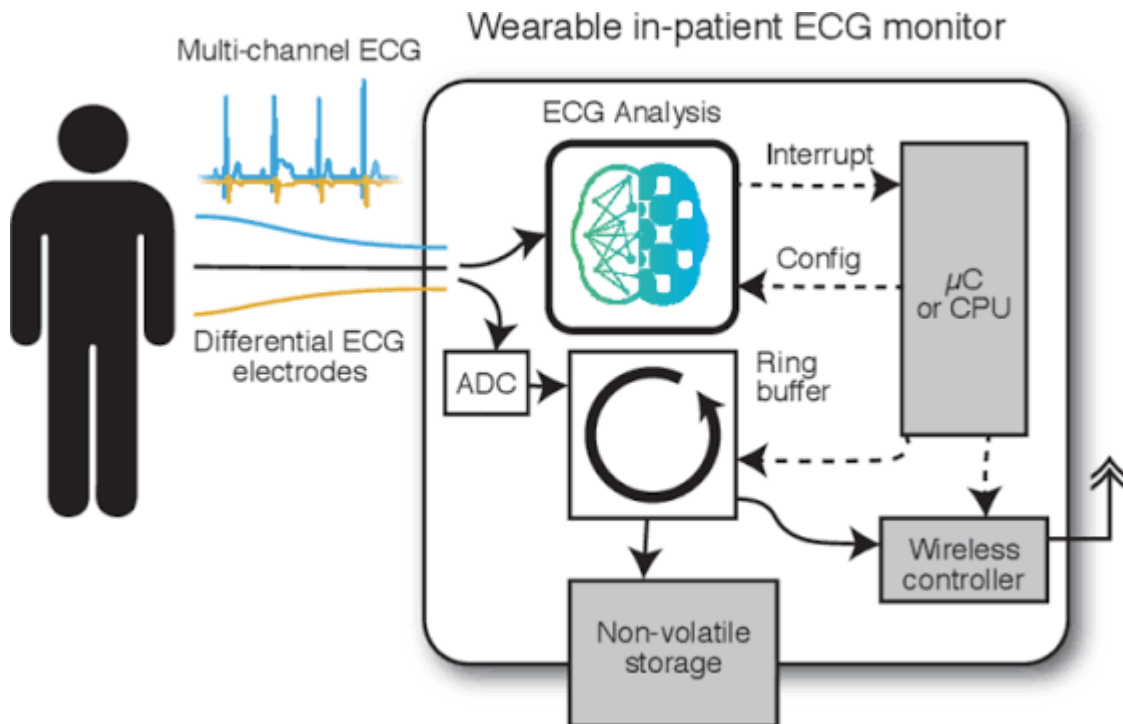
Proposed Improvements

- Increase the number of hidden layers.
- Increase the number of neurons where necessary.
- Use **ReLU** activation to improve convergence speed.
- Apply **Leaky ReLU** to avoid dead neuron problems.
- Introduce **Batch Normalization** for stable learning.
- Add **Dropout** layers to reduce overfitting.
- Use **Softmax** activation for multi-class face recognition.
- Train using a larger and more diverse face dataset.

Benefits

- Learns detailed facial representations.
- Higher recognition accuracy.

- Better performance under varying environmental conditions.
- Faster model convergence.
- Reduced overfitting.
- Improved generalization to unseen faces.



Applications

- Smartphone face unlock
- Airport security systems
- Attendance management
- Criminal identification
- Smart surveillance

Expected Result

The deep neural network extracts robust facial features and significantly improves recognition accuracy even under challenging environmental conditions. The upgraded architecture provides reliable face identification suitable for real-world security applications.

4.Restricted Boltzmann Machines (RBMs), Unsupervised Training of Neural Networks

A streaming service wants to recommend movies to users based on their viewing history without using labeled data. Explain how RBMs can be applied to build the recommendation engine.

Problem Statement

Online streaming platforms contain thousands of movies and television shows. Users often struggle to discover content matching their interests. Since explicit labels indicating user preferences are unavailable, an unsupervised recommendation technique is required to learn hidden viewing patterns automatically.

Proposed Solution

A Restricted Boltzmann Machine (RBM) is an unsupervised neural network capable of learning latent relationships between users and movies. It contains a visible layer representing user ratings and a hidden layer representing hidden preference features. During training, the RBM discovers meaningful viewing patterns and predicts ratings for unseen movies.

System Workflow

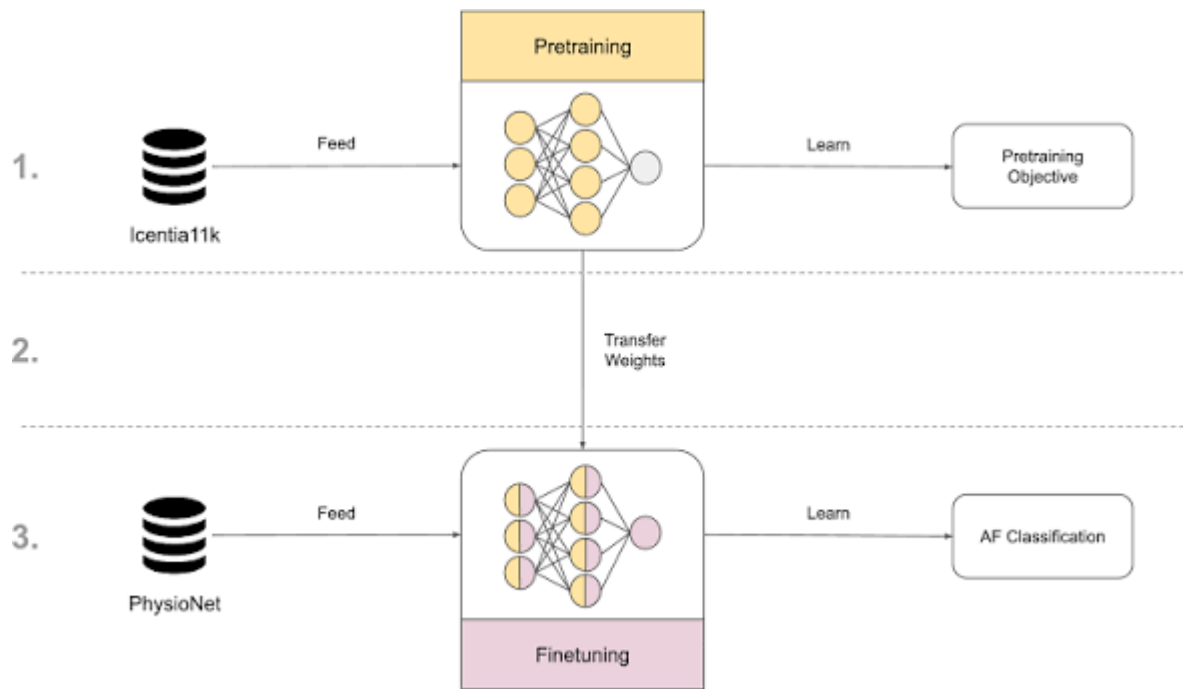
- Collect user viewing history.
- Create a user-item interaction matrix.
- Train the RBM using Contrastive Divergence.
- Learn hidden user preference features.
- Predict missing movie ratings.
- Recommend highly rated unseen movies.

Hidden Features Learned

- Favorite movie genres.
- Preferred actors and directors.
- Viewing frequency.
- Weekend and holiday watching habits.
- Language preferences.
- Similar user interests.

Advantages

- Requires no labeled data.
- Learns personalized recommendations.
- Handles sparse datasets efficiently.
- Improves customer engagement.
- Increases watch time.
- Enhances subscription retention.



Applications

- Netflix
- Amazon Prime Video
- Disney+
- Spotify music recommendations
- Online shopping recommendation systems

Expected Result

The RBM-based recommendation system predicts user interests with high accuracy and delivers personalized movie suggestions, improving user satisfaction and increasing platform engagement.

5.Deep Learning Applications, Representation Learning

A manufacturing company wants to detect defective products from sensor readings collected every second. Design a deep learning solution using concepts from representation learning and neural networks.

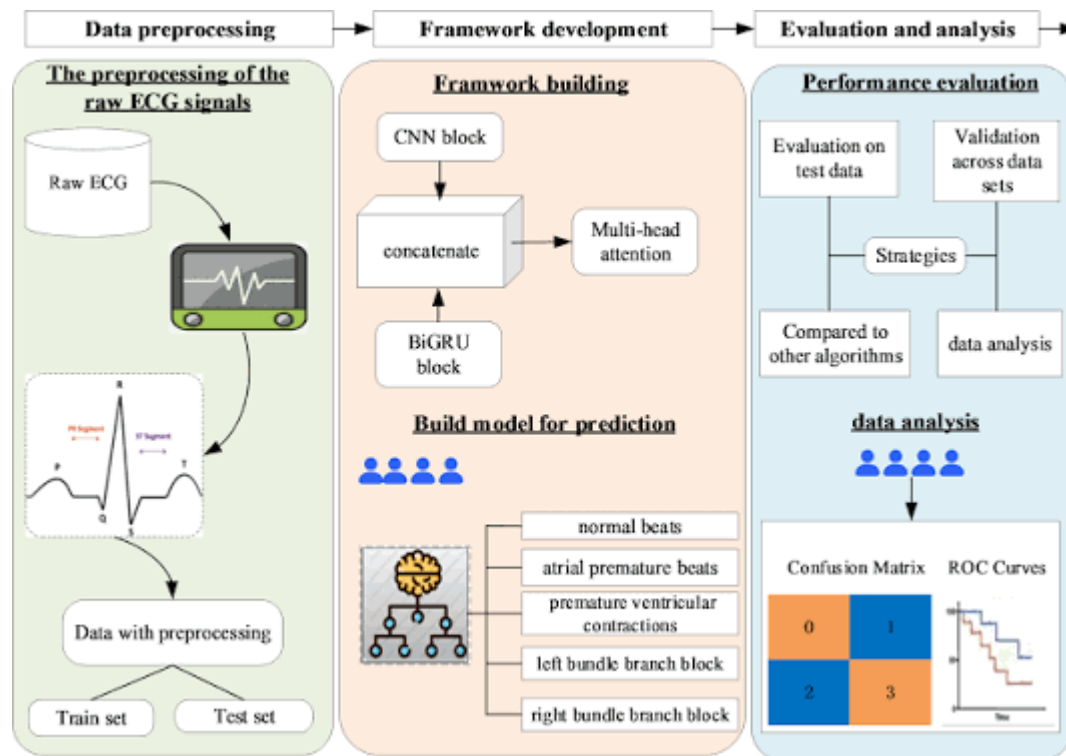
Problem Statement

Modern manufacturing plants generate continuous sensor data from machines measuring temperature, vibration, pressure, current, humidity, and rotational speed. Detecting defective products manually is inefficient and prone to human error. Therefore, an intelligent deep

learning system is required for automatic defect detection.

Proposed Deep Learning Framework

A **Representation Learning model using an Autoencoder** learns the normal operational characteristics of manufacturing equipment. Instead of manually designing features, the model automatically extracts meaningful latent representations from sensor readings and detects anomalies based on reconstruction error.



System Workflow

- Collect real-time sensor readings.
- Remove noisy and inconsistent values.
- Normalize sensor measurements.
- Train the autoencoder on normal production data.
- Learn compressed machine behavior representations.
- Reconstruct sensor readings.
- Calculate reconstruction error.

- Detect abnormal products exceeding the threshold.
- Notify operators immediately.

Key Features

- Automatic feature learning.
- Continuous production monitoring.
- Real-time anomaly detection.
- Predictive maintenance support.
- Intelligent quality inspection.
- Reduced manual intervention.

Industrial Benefits

- Improved product quality.
- Reduced manufacturing defects.
- Lower maintenance costs.
- Increased machine availability.
- Higher production efficiency.
- Reduced operational downtime.
- Enhanced customer satisfaction.
- Better quality assurance.

Applications

- Automobile manufacturing
- Electronics assembly
- Pharmaceutical production
- Food processing industries
- Smart factories (Industry 4.0)

Expected Result

The proposed intelligent manufacturing system continuously monitors production equipment and identifies defective products at an early stage. By using representation learning and autoencoders, the system improves quality control, minimizes production losses, reduces maintenance expenses, and enhances overall manufacturing efficiency.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand